

Use-Case Flow Specification

Academic Elective Bidding & Allocation System

Use Case ID: UC-03 | Use Case Name: Submit Bids

Actor(s)	Student (primary); Academic Registrar (secondary — receives submitted bid data)
Trigger	Student selects preferred electives and submits a bid for the current bidding round.
Includes	Check Prerequisites (mandatory, always performed — FR-001)
Extends	Check Timetable Conflicts (conditional, invoked only when a clash is detected — FR-003)
Related Requirements	FR-001, FR-003, NFR-001, NFR-002

1. Preconditions

1. The Student is authenticated and logged into the Academic Elective Bidding & Allocation System (session established via role-based authorization, per NFR-002).
2. The current academic term's bidding window is open (the current date and time fall within the configured start and end of the bidding period).
3. The list of Available Electives for the term has been published by the Academic Registrar (see the "View Available Electives" use case).
4. The Student's bidding credit balance is initialized at exactly 100 credits for the current round, per FR-001.
5. The Student has not already submitted a final bid for this bidding round, or the system permits revision of an existing bid before the window closes.

2. Postconditions

Success Guarantee

- The Student's ranked elective choices, with credits allocated per choice, are recorded as a submitted bid, timestamped and linked to the Student's ID.
- The total allocated credits are confirmed to be less than or equal to 100 and are logged as consumed from the Student's credit balance (FR-001).
- No unresolved timetable clash exists in the submitted bid (FR-003).
- The bid record is made available to the Academic Registrar for the subsequent "Allocate Electives" process.

- The Student receives on-screen confirmation of successful submission.

Minimal Guarantee (on failure)

- No partial or corrupted bid record is saved; system state — including the Student's 100-credit balance — remains unchanged from before the attempt.

3. Main Success Scenario

1. The Student navigates to the bid submission screen after reviewing the list of Available Electives.
2. The system displays the electives eligible for bidding, along with available seats and the Student's remaining credit balance out of a strict 100-credit limit (FR-001).
3. The Student selects and ranks their preferred electives and allocates credits to each choice, such that the sum of allocated credits does not exceed the 100-credit budget (FR-001).
4. The Student submits the bid form.
5. The system performs the included Check Prerequisites use case for each selected elective, confirming the Student meets all prerequisite requirements (FR-001).
6. The system validates the bid data: total allocated credits must not exceed 100, no duplicate rankings are allowed, and the elective count is within permitted limits. Any submission where allocated credits exceed 100 is rejected outright (FR-001).
7. The system checks the bid against the Student's existing timetable for schedule clashes (see Alternate Flow A1, below).
8. The system saves the bid record and associates it with the Student's profile and the current bidding round.
9. The system displays a confirmation message summarizing the submitted bid, including the final credit allocation and remaining balance.

4. Alternate Flow A1 — Timetable Conflict Detected

Extends: Check Timetable Conflicts

Trigger point: Occurs at Step 7 of the Main Success Scenario, once credit validation and prerequisite checks have passed and the system evaluates the bid against the Student's existing schedule.

1. The system detects that one or more selected electives overlap in schedule with another elective already on the Student's timetable, invoking the extension point Check Timetable Conflicts.

2. The system flags the conflicting elective(s) and displays a hard-block error to the Student, identifying the specific time-slot clash.
3. Per FR-003, the system does not permit provisional or conflict-flagged submission under any circumstance; the bid submission is blocked and no bid record is created.
4. The Student is required to revise their elective selection and/or credit allocation to remove the detected conflict.
5. The flow resumes at Step 3 of the Main Success Scenario. The Student cannot proceed to Step 8 (save) until the system re-validates the revised bid with zero timetable conflicts.

Resumption: Flow rejoins the Main Success Scenario at Step 3. There is no path from Alternate Flow A1 directly to Step 8; a conflict-free re-validation is mandatory, since FR-003 imposes a hard block with no override permitted.

5. Non-Functional Requirements Mapping

The following non-functional requirements constrain how UC-03 must be implemented and are binding on every step of the Main Success Scenario and Alternate Flow A1.

NFR-001 — Performance

All interactive steps of UC-03 (Steps 1 through 4, the Step 7 feedback, and the Alternate Flow A1 warning) shall return sub-second user-facing responses under normal load. The backend allocation and validation engine invoked during bid submission shall be capable of processing 5,000 concurrent student bids and resolving all associated schedule conflicts in under 30 seconds during the final allocation run.

NFR-002 — Security & Role-Based Access Control (RBAC)

All data transmitted during Steps 1 through 9 — credit allocation, elective selections, prerequisite results, and timetable data — shall be encrypted in transit (TLS) and at rest. Access to UC-03 is restricted via strict role-based access control: only authenticated Students may submit bids on their own behalf, and only the Academic Registrar role may read aggregated bid records for allocation. No role may bypass the 100-credit limit (FR-001) or the timetable hard block (FR-003) through direct data access.

6. Requirements Traceability Matrix

UC-03 Step	FR-001	FR-003	NFR-001	NFR-002
Step 2 — Display credit balance	X		X	
Step 3 — Allocate ≤ 100 credits	X			X

UC-03 Step	FR-001	FR-003	NFR-001	NFR-002
Step 5 — Check Prerequisites (include)	X			X
Step 6 — Validate credit total ≤ 100	X		X	X
Step 7 — Timetable conflict check		X	X	
Step 8 — Save bid record	X	X	X	X
Step 9 — Confirmation display	X		X	
Alt. Flow A1 — Hard block on conflict		X		X